Bapuji Educational Association(R)
Bapuji Institute of Engineering and Technology
Davanagere-577004, Karnataka
An Autonomous Institute Affiliated to Visvesvaraya Technological University, Belagavi, Karnataka.
Approved by AICTE, New Delhi, Accredited by NAAC with 'A' grade and NBA

IDP PROJECT REPORT

BIDPRJ218

BE Programs

Common to All Branches

On

"AI-Driven Smart Energy Saving Advisor for Sustainable Electricity Consumption"

Submitted by:

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Under the Mentorship of

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2025-2026

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CERTIFICATE

Certified that the project work entitled "**AI-Driven Smart Energy Saving Advisor for Sustainable Electricity Consumption**" for IDP Lab (BIDPRJ218),carried out by MANYA J JAIN USN-4BD25CS097 Bonafide students of Second Semester in partial fulfillment for the award of Bachelor of Engineering of Visvesvaraya Technological University, Belagavi during the academic year 2025-2026.

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Vision and Mission of the Institute

Vision

"To be a center of excellence recognized nationally and **internationally,** in distinctive areas of engineering education and research, based on a culture of **innovation** and invention"

Mission

"BIET contributes to the growth and **development** of its students by imparting a broad-based engineering education and empowering them to be successful in their chosen field by inculcating in them positive approach, **leadership** qualities and **ethical** values"

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ABSTRACT

This project proposes a Smart Energy Saving Advisor system designed to monitor electricity consumption and provide intelligent energy-saving suggestions in real time. The system integrates various sensors such as the ACS712 current sensor for measuring power usage, DHT11 sensor for temperature monitoring, and LDR sensor for light intensity detection using an Arduino UNO microcontroller. The collected data is continuously processed and transmitted to a web-based dashboard where users can monitor energy consumption, temperature, humidity, and light intensity in real time. Based on the monitored energy usage patterns, the system generates AI-based suggestions such as reducing unnecessary appliance usage, switching off lights, and optimizing air-conditioner usage to improve energy efficiency. This project aims to promote sustainable energy management, reduce electricity wastage, and increase awareness about smart energy conservation using IoT and simple artificial intelligence techniques.

TABLE OF CONTENTS

CHAPTER – 1
INTRODUCTION AND OBJECTIVES

1.1 INTRODUCTION

Energy conservation has become one of the most important challenges in modern society due to increasing electricity consumption and growing environmental concerns. The widespread use of electrical appliances in homes, offices, industries, and educational institutions has significantly increased power demand. However, lack of awareness regarding energy-efficient practices often results in unnecessary electricity wastage, increased operational costs, and environmental impact. Therefore, intelligent systems capable of monitoring energy consumption and providing recommendations for optimization are becoming increasingly important.

The Smart Energy Saving Advisor is an IoT and Artificial Intelligence based energy monitoring system designed to analyse electricity consumption and provide intelligent recommendations for reducing energy wastage. The project combines hardware and software technologies to create an efficient, user-friendly, and sustainable energy management solution.

The hardware section consists of Arduino UNO, ACS712 current sensor, DHT11/ temperature and humidity sensor, LDR sensor, WEB DASHBOARD display, LEDs, resistors, and supporting components. The ACS712 sensor monitors the current consumed by electrical appliances, while the DHT11/ sensor measures environmental conditions such as temperature and humidity. The LDR sensor detects surrounding light intensity to determine whether artificial lighting is necessary. The collected sensor data is processed by the Arduino UNO and transmitted to the Smart Energy Saving Advisor Web Dashboard for real-time monitoring and analysis.

In addition to the hardware setup, the project includes a web-based monitoring platform known as the Smart Energy Advisor Web Platform. The dashboard provides a centralized interface for displaying live sensor readings, graphical energy usage statistics, and AI-based recommendations. The collected data from the hardware system is transmitted to the dashboard, allowing users to monitor energy consumption remotely through computers and mobile devices.

The website analyses real-time data and provides intelligent recommendations such as switching off unused appliances, reducing electricity usage during peak hours, and optimizing air-conditioner operation according to room temperature. The integration of hardware monitoring and web-based analytics enables users to make informed decisions regarding electricity consumption and energy conservation.

The project follows a practical implementation approach where the circuit was initially tested using Wokwi simulation and later implemented using actual hardware components. The successful integration of sensors, microcontroller, WEB DASHBOARD display, and Web Dashboard demonstrates the practical application of IoT and AI technologies for sustainable energy management.

The proposed system is cost-effective, scalable, and suitable for deployment in homes, offices, educational institutions, and small industries. It promotes awareness regarding responsible electricity usage and contributes toward environmental sustainability through intelligent energy management practices.

The Smart Energy Saving Advisor platform includes a secure login system that allows users to access personalized energy reports and recommendations. The website analyzes energy usage patterns and provides AI-generated suggestions, estimated cost savings, and carbon footprint reduction insights to encourage sustainable energy consumption.

1.2 OBJECTIVES

The major objectives of the Smart Energy Saving Advisor project are:

1. To design and develop an IoT-based smart energy monitoring system.
2. To monitor real-time electricity consumption using ACS712 current sensor.
3. To monitor environmental parameters such as temperature, humidity, and light intensity.
4. To process sensor data using Arduino UNO microcontroller.
5. To display real-time sensor readings on a web-based dashboard.
6. To integrate hardware sensors with the Smart Energy Advisor Web Platform website.
7. To provide remote monitoring of energy consumption through a web interface.
8. To generate AI-based energy-saving recommendations.
9. To reduce unnecessary electricity wastage and improve energy efficiency.
10. To promote sustainable energy management and environmental awareness.
11. To develop a low-cost and user-friendly smart energy monitoring solution.
12. To provide a scalable platform for future smart home and automation applications.
13. To provide secure user authentication and account management.
14. To promote carbon footprint reduction and sustainability.

CHAPTER – 2

LITERATURE SURVEY

The increasing demand for electricity and the growing need for sustainable energy management have encouraged researchers to develop intelligent energy monitoring systems. Traditional energy monitoring methods mainly focused on measuring electricity consumption without providing users with recommendations for energy optimization. With advancements in Internet of Things (IoT), Artificial Intelligence (AI), cloud computing, and web technologies, modern energy management systems have become more efficient, intelligent, and user-friendly.

In 2024, researchers developed smart energy monitoring systems that combined IoT sensors with AI-based analytics to monitor electricity consumption in real time. These systems collected data from various sensors and generated intelligent recommendations to reduce energy wastage. Web-based dashboards were also introduced to display energy usage statistics and graphical analysis, enabling users to monitor their electricity consumption remotely. The studies concluded that intelligent monitoring systems significantly improve energy efficiency and support sustainable development.

In 2023, cloud-based energy management systems gained popularity due to their ability to provide remote monitoring and data analysis. Researchers integrated sensors, microcontrollers, and cloud platforms to collect and store real-time electricity consumption data. The collected information was analysed to identify inefficient energy usage patterns and generate recommendations for optimization. These systems improved accessibility and helped users make informed decisions regarding electricity consumption.

In 2022, low-cost IoT solutions based on Arduino and ESP32 microcontrollers were widely used for smart energy monitoring applications. Researchers implemented systems using current sensors, temperature sensors, and light sensors to monitor household and industrial energy usage. The collected data was displayed through Web Dashboards and mobile applications. The studies showed that continuous monitoring and real-time feedback help users reduce unnecessary energy consumption and improve energy efficiency.

In 2021, researchers focused on integrating IoT platforms with web-based dashboards for real-time energy monitoring. These systems monitored parameters such as voltage, current, power consumption, temperature, and environmental conditions. Graphical visualization tools were used to present energy usage trends, making it easier for users to understand and optimize their consumption patterns. The research highlighted the importance of remote accessibility and data visualization in modern energy management systems.

Artificial Intelligence played a significant role in energy management systems during 2020. Researchers developed machine learning models capable of predicting energy consumption patterns and generating intelligent recommendations based on historical and real-time data. These systems improved decision-making capabilities and helped users optimize appliance usage. The studies demonstrated that AI-based energy management systems are more effective than conventional monitoring systems in reducing electricity wastage.

In 2019, IoT-based smart energy meters were extensively used in residential and industrial applications. These systems utilized sensors and wireless communication technologies to monitor electrical loads remotely. Researchers found that real-time monitoring and automated reporting increased user awareness regarding electricity consumption and encouraged energy-saving practices. The studies emphasized the importance of intelligent monitoring systems in achieving sustainable energy utilization.

From the literature survey, it is evident that the integration of IoT, Artificial Intelligence, and web technologies has transformed traditional energy monitoring systems into intelligent energy management solutions. Most existing systems focus on real-time monitoring, data analysis, and energy optimization. The proposed **Smart Energy Saving Advisor** project follows a similar approach by integrating Arduino UNO, ACS712 current sensor, DHT11 sensor, LDR sensor, WEB DASHBOARD display, and the **Smart Energy Advisor Web Platform** website. The system provides real-time monitoring, graphical visualization, and AI-based energy-saving recommendations, making it an effective and sustainable solution for modern energy management.

CHAPTER – 3

MATERIALS AND METHODOLOGY

The Smart Energy Saving Advisor is an IoT and AI-based system developed to monitor energy consumption and provide intelligent energy-saving recommendations. The system integrates hardware components with a web-based dashboard to enable real-time monitoring and analysis of electricity usage.

The project was developed in two phases. Initially, the circuit was designed and tested using the Wokwi simulation platform to verify the functionality of all sensors and components. After successful simulation, the circuit was implemented using actual hardware components and connected to the Smart Energy Advisor Web Platform for real-time monitoring.

The hardware setup consists of Arduino UNO, ACS712 current sensor, DHT11 temperature and humidity sensor, LDR sensor, WEB DASHBOARD display, LEDs, resistors, breadboard, and jumper wires. The ACS712 sensor measures electrical current consumption, the DHT11 sensor measures temperature and humidity, and the LDR sensor detects ambient light intensity.

The Arduino UNO acts as the central controller of the system. It collects data from all sensors, processes the readings, and displays the information on the WEB DASHBOARD display. The processed data is also transmitted to the Smart Energy Advisor Web Platform website, where it is analysed and presented through graphical representations and real-time monitoring panels.

The Smart Energy Advisor Web Platform serves as the software interface of the project. It allows users to monitor sensor readings remotely through computers and mobile devices. The dashboard also generates energy-saving recommendations based on predefined conditions and energy usage patterns.

Website Features

The Smart Energy Saving Advisor website provides a user-friendly platform for monitoring energy usage and receiving personalized recommendations. The platform includes a secure login system that allows users to create accounts and access personalized energy reports. The website also provides AI-powered recommendations, estimated electricity cost savings, and carbon footprint analysis to promote sustainable energy consumption.

Main Features:

- User Registration and Login
- Personalized Dashboard
- AI-Based Energy Recommendations
- Energy Savings Analysis
- Carbon Footprint Monitoring
- Responsive Web Interface

METHODOLOGY

The working methodology of the Smart Energy Saving Advisor system is explained through the following steps:

Step 1- Sensor Data Collection: The ACS712 current sensor measures electrical current consumption, the DHT11 sensor measures temperature and humidity, and the LDR sensor measures ambient light intensity. These sensors continuously collect real-time data from the environment.

Step 2- Data Processing: The collected sensor readings are transmitted to the Arduino UNO microcontroller. The Arduino processes the received data using Embedded C programming and prepares it for display and transmission.

Step 3- Dashboard Monitoring: The processed sensor values are transmitted to the Smart Energy Saving Advisor dashboard and displayed in real time, enabling users to monitor electricity consumption and environmental conditions locally.

Step 4- Data Transmission: The Arduino transmits the processed sensor readings to the Smart Energy Advisor Web Platform through communication interfaces, enabling remote monitoring and visualization.

Step 5- Website Analysis: The Smart Energy Advisor Web Platform receives the sensor data and displays it in the form of readings, charts, and graphical reports. The dashboard provides a clear visualization of energy consumption patterns.

Step 6- AI-Based Recommendation Generation: The dashboard analyses the received data using predefined thresholds and AI-based logic. Based on the observed energy usage patterns, intelligent recommendations are generated to reduce electricity wastage and improve efficiency.

Step 7- User Action and Energy Optimization: The user follows the generated recommendations to optimize electricity consumption, resulting in improved energy efficiency and sustainable energy management.

SYSTEM ARCHITECTURE

The overall system architecture consists of the following modules:

1. Data Acquisition Module (Sensors)
2. Processing Module (Arduino UNO)
3. Display Module (WEB DASHBOARD Display)
4. Communication Module
5. Smart Energy Advisor Web Platform Website
6. AI-Based Recommendation Module
7. User Monitoring and Decision-Making Module

The flow of information in the system is:

Sensors → Arduino UNO → Web Dashboard → AI Analysis → Energy Saving Suggestions

3.1 HARDWARE REQUIREMENTS SPECIFICATIONS

The hardware components used in the Smart Energy Saving Advisor project are listed below:

Sl. No.	Hardware Component	Purpose
1	Arduino UNO	Main microcontroller for processing sensor data
2	ACS712 Current Sensor	Measures electrical current consumption
3	DHT11 Sensor	Measures temperature and humidity
4	LDR Sensor	Measures ambient light intensity
6	LED	Status indication
7	Resistors	Circuit protection and voltage division
8	Breadboard	Circuit prototyping
9	Jumper Wires	Component interconnections
10	USB Cable	Programming and power supply

3.2 SOFTWARE REQUIREMENTS SPECIFICATION

The software tools and technologies used in the project are listed below:

Sl. No.	Software Tool	Purpose
1	Arduino IDE	Program development and code uploading
2	Wokwi Simulator	Circuit simulation and testing
3	Embedded C	Arduino programming language
4	Smart Energy Advisor Web Platform	Real-time monitoring and analysis
5	Lovable Platform	Dashboard development
6	HTML, CSS, JavaScript	Website interface design
7	AI-Based Logic	Generation of energy-saving recommendations
8	Windows Operating System	Development environment
9	Authentication Module	User Login and Registration
10	AI Recommendation Engine	Personalized Suggestions
11	Savings Analysis Module	Cost Saving Estimation
12	Carbon Impact Module	Environmental Analysis

The successful integration of hardware sensors, Arduino UNO, WEB DASHBOARD display, and Smart Energy Advisor Web Platform demonstrates the practical implementation of IoT and Artificial Intelligence technologies for smart energy management and sustainability.

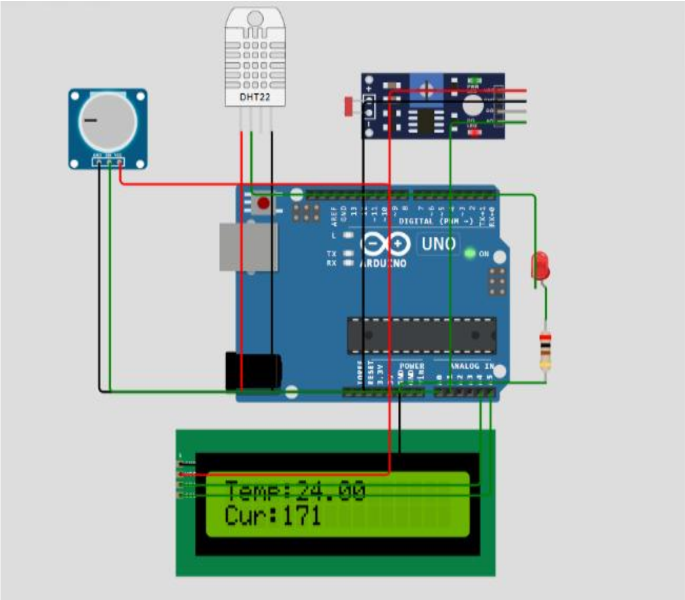


Figure 3.1: Wokwi Simulation of Smart

Energy Saving Advisor

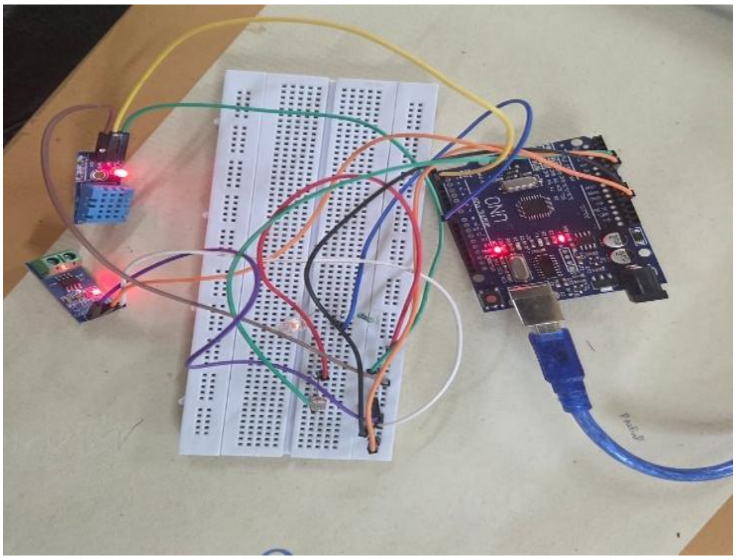


Figure 3.2: Hardware Setup

CHAPTER 4
RESULTS AND DISCUSSION

The Smart Energy Saving Advisor system was successfully designed, implemented, and tested using both simulation and physical hardware components. Initially, the complete circuit was developed and verified using the Wokwi simulation platform to ensure proper functionality of all sensors and modules. After successful simulation, the circuit was physically assembled using Arduino UNO, ACS712 current sensor, DHT11 sensor, LDR sensor, WEB DASHBOARD display, LEDs, resistors, breadboard, and jumper wires.

The hardware setup successfully collected real-time environmental and electrical data. The ACS712 current sensor continuously monitored the electrical current consumed by connected loads and provided accurate current readings. The DHT11 sensor effectively measured temperature and humidity conditions, while the LDR sensor monitored ambient light intensity. The Arduino UNO processed all sensor readings and transmitted them to the Smart Energy Saving Advisor dashboard, where the values were displayed in real time.

The collected sensor data was further integrated with the Smart Energy Advisor Web Platform website. The dashboard successfully displayed real-time values of current consumption, temperature, humidity, and light intensity. Users were able to monitor the system remotely through the web interface using computers and mobile devices. The dashboard also provided graphical visualization of the collected data, making it easier to analyse energy consumption patterns.

The AI-based recommendation module analysed the received sensor values and generated intelligent suggestions for energy conservation. When the system detected high current consumption, recommendations were generated to reduce appliance usage or switch off unnecessary devices. Similarly, when sufficient daylight was available, the dashboard suggested reducing artificial lighting to conserve electricity. Temperature readings were also analysed to recommend efficient usage of cooling appliances.

The implementation demonstrated the successful integration of IoT sensors, embedded systems, and web technologies for sustainable energy management. The system effectively monitored energy consumption and provided useful recommendations that can help users reduce electricity wastage and improve energy efficiency.

OBSERVED RESULTS

The following observations were obtained during system testing:

1. Real-time sensor data was successfully collected and processed.
2. The ACS712 sensor accurately monitored electrical current consumption.
3. The DHT11 sensor provided stable temperature and humidity readings.
4. The LDR sensor successfully detected changes in ambient light intensity.
5. The WEB DASHBOARD display showed live sensor readings without noticeable delay.
6. Sensor data was successfully transmitted to the Smart Energy Advisor Web Platform.
7. The dashboard displayed real-time monitoring information and graphical analysis.
8. AI-based recommendations were generated according to sensor conditions.
9. The system operated reliably during continuous monitoring.
10. The integration between hardware and website functioned successfully.
11. User login and authentication functioned successfully.
12. AI-based personalized recommendations were generated correctly.
13. The system estimated electricity cost savings effectively.
14. Carbon footprint analysis was displayed successfully.
15. Users were able to access personalized dashboards after login.

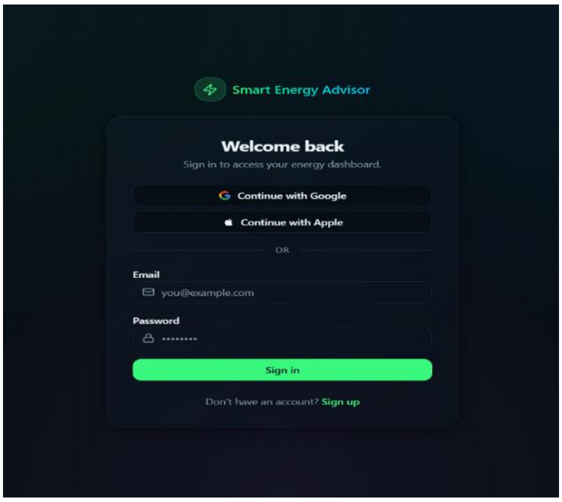
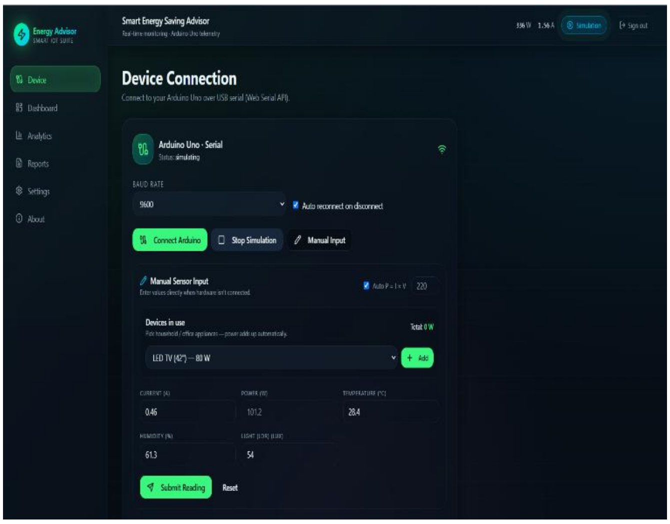


Figure 3.3: Login Page

Figure 3.4: Dashboard Home Page



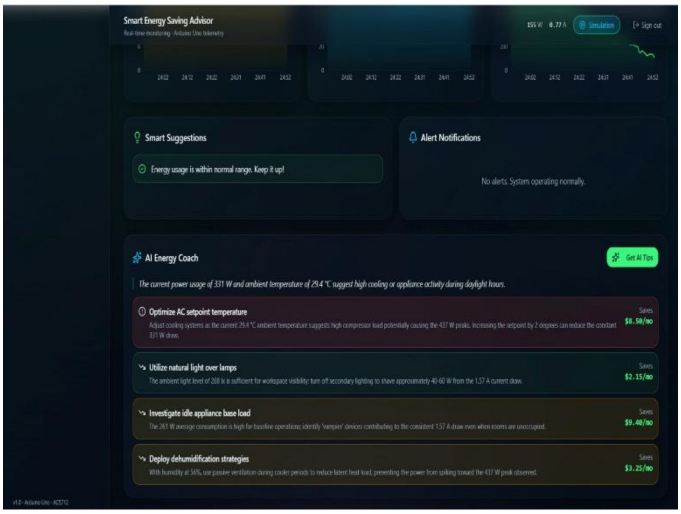


Figure 3.5: Device Connection

Figure 3.6: AI suggestions

DISCUSSION

The obtained results indicate that the Smart Energy Saving Advisor system is capable of providing efficient real-time energy monitoring and intelligent decision support. The integration of sensors with Arduino UNO enabled accurate data collection and processing. The Smart Energy Advisor Web Platform enhanced the functionality of the system by providing remote accessibility, visualization, and user interaction.

The implementation of user authentication improved security and personalization within the platform. The AI recommendation module generated customized energy-saving suggestions based on user energy consumption patterns. The savings analysis feature helped users understand the potential reduction in electricity bills, while the carbon footprint analysis promoted awareness regarding environmental sustainability.

Compared to conventional energy monitoring systems, the proposed system not only measures energy-related parameters but also provides intelligent recommendations for reducing energy consumption. This feature improves user awareness regarding energy conservation and encourages sustainable energy usage practices.

The use of low-cost components such as Arduino UNO, ACS712, DHT11, and LDR sensors makes the system economically feasible for residential, educational, and small-scale commercial applications. Furthermore, the modular architecture of the system allows future expansion through cloud integration, machine learning algorithms, automated appliance control, and smart home automation technologies.

The successful implementation of both simulation and real hardware validates the reliability and effectiveness of the proposed system. The project demonstrates how IoT and Artificial Intelligence technologies can be utilized to address modern energy management challenges and promote sustainability.

ADVANTAGES OF THE PROPOSED SYSTEM

1. Real-time monitoring of electricity consumption.
2. Intelligent energy-saving recommendations.
3. Integration of hardware and web technologies.
4. Remote monitoring through a Web Dashboard.
5. Low-cost and user-friendly implementation.
6. Improved energy efficiency and conservation.
7. Scalable architecture for future enhancements.
8. Support for sustainable energy management.
9. Easy deployment in homes, offices, and educational institutions.
10. Increased awareness regarding responsible electricity usage.
11. Secure user authentication and account management.
12. Personalized AI-based recommendations.
13. Energy cost savings estimation.
14. Carbon footprint monitoring and sustainability awareness.

The overall results confirm that the Smart Energy Saving Advisor system successfully achieves its objective of monitoring energy consumption and providing intelligent recommendations for efficient and sustainable electricity usage.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

The Smart Energy Saving Advisor project was successfully designed and implemented as an IoT and Artificial Intelligence based energy monitoring system. The project effectively integrated hardware components such as Arduino UNO, ACS712 current sensor, DHT11 sensor, LDR sensor, WEB DASHBOARD display, and supporting electronic components with the Smart Energy Advisor Web Platform website to create a complete smart energy management solution.

The system continuously monitored electrical current consumption, temperature, humidity, and light intensity in real time. The collected sensor data was processed by the Arduino UNO and displayed on the WEB DASHBOARD display as well as the Web Dashboard. The Smart Energy Advisor Web Platform successfully visualized the sensor readings and generated AI-based energy-saving recommendations based on user energy consumption patterns.

The platform also provided secure user authentication, personalized energy reports, electricity cost-saving estimations, and carbon footprint analysis. These features transformed the project from a simple monitoring system into a comprehensive smart energy management platform.

The project demonstrated how IoT and Artificial Intelligence technologies can be utilized to improve energy efficiency and reduce electricity wastage. The generated recommendations helped users identify unnecessary power consumption and encouraged the adoption of energy-efficient practices. The integration of hardware monitoring and web-based analytics enhanced accessibility, usability, and decision-making capabilities.

The successful implementation of both simulation and physical hardware validated the reliability and effectiveness of the proposed system. The project achieved its primary objective of developing a smart, low-cost, and user-friendly energy monitoring solution that promotes sustainable energy management and environmental awareness.

Overall, the Smart Energy Saving Advisor provides an efficient platform for monitoring electricity consumption, generating intelligent recommendations, and supporting sustainable energy conservation practices in homes, offices, educational institutions, and small-scale industries.

5.2 FUTURE SCOPE

Although the developed system successfully performs real-time monitoring and recommendation generation, several improvements can be incorporated in future versions to enhance its functionality and performance.

1. Integration of cloud computing platforms for secure data storage and remote access.
2. Implementation of advanced Machine Learning algorithms for predictive energy consumption analysis.
3. Automatic control of electrical appliances based on AI-generated recommendations.
4. Development of a dedicated Android and iOS mobile application.
5. Integration with smart home automation systems.
6. Real-time energy consumption alerts and notifications.
7. Voice assistant integration using technologies such as Google Assistant and Alexa.
8. Renewable energy monitoring for solar and wind energy systems.
9. Advanced data analytics and personalized user recommendations.
10. Multi-user monitoring and management capabilities for large buildings and industries.
11. Integration of wireless communication technologies such as Wi-Fi, Bluetooth, and GSM.
12. Implementation of smart billing and electricity cost estimation features.

The future enhancements will transform the system into a more intelligent, autonomous, and scalable smart energy management platform capable of addressing the growing challenges of energy conservation and sustainability.

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